Antonio P. Sberna

Research Fellow of Structural Engineering

Department of Engineering and Geology
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Personal information

Language skills	Italian (native), English (fluent)
Citizenship	Italian
Professional qualification	Professional licensed engineer qualified on 2nd national exam 2020 Registered Professional Engineer in Italy (Enna, numb. 856)

Education

Nov 2020 – Sep 2024 **PhD in Civil and Environmental Engineering**, *Politecnico di Torino*
Ph.D. thesis: *Engineered design frameworks for the seismic retrofitting of existing structures*
Advisors: Prof. Giuseppe Marano and Prof. Fabio Di Trapani

Oct 2017 – Jul 2020 **MS in Civil Engineering**, *Politecnico di Torino*
Master thesis (in english): *Optimal seismic retrofitting of reinforced concrete buildings by steel-jacketing using a genetic algorithm-based framework*
Advisors: Prof. Fabio Di Trapani and Prof. Giuseppe Marano

Oct 2013 – Apr 2017 **BS in Civil and Environmental Engineering**, *Università degli studi di Catania*
Bachelor thesis (in italian): *Influenza dello sforzo normale sul collasso plastico di strutture intelaiate* (*Effect of axial load on the plastic collapse of frame structures*)
Advisor: Prof. Annalisa Greco

Academic positions

Feb 2026 – present Postdoctoral Research Fellow
University of Chieti-Pescara (Italy), Department of Engineering and Geology

Feb 2024 – Feb 2026 Postdoctoral Research Fellow
Politecnico di Torino (Italy), Department of Structural, Geotechnical and Building Engineering

Visiting research experiences

Dec 2022 – Dec 2023 *University of California San Diego* (United States)
Research topic: *Seismic retrofit optimization of non-ductile RC buildings using soft-computing techniques within the framework of probabilistic performance-based earthquake engineering.*
Advisor: Prof. J. P. Conte

Publications in international journals

1. Zhou Y., Ye F., Shi D., Yang Z., *Sberna A.P.*, Xiao Y., Demartino C. "Characterizing bamboo aggregate foamed concrete for sustainable construction: thermal, acoustic, mechanical, and environmental aspects". *Materials and Structures*, 2026; 59:145.
2. Sadeghzadeh S., *Sberna A.P.*, Di Trapani F., Demartino C. "Enhancing seismic site response analyses: tuning soil properties via genetic algorithms and Bayesian model updating from downhole array data". *Soil Dynamics and Earthquake Engineering*, 2026; 200:109773.
3. Wang J., Demartino C., *Sberna A.P.*, Jiang L., Usmani A. "Enhanced Compressive Performance and Energy Absorption in SLM-Fabricated 316L Arrowhead Auxetics via Tendon and Stuffer Geometry Modification". *Composite Structures*, 2025; 374:119710.
4. Di Trapani F., Di Benedetto M., *Sberna A.P.*, Camata G. "Local Shear Demand Correction Model for the Analysis of Infilled Frames Using Equivalent Struts". *Journal of Structural Engineering*, 2025; 151(12):04025208.
5. *Sberna A.P.*, Deb A., Di Trapani F., Conte J.P. "Reliability-based seismic retrofitting design methodology for non-ductile reinforced concrete frame structures". *Probabilistic Engineering Mechanics*, 2025; 82:103818.
6. *Sberna A.P.*, Di Benedetto M., Di Trapani F. "Engineered frameworks for sustainable seismic retrofitting design: a state-of-the-art review". *Bulletin of Earthquake Engineering*, 2025; 23:4683–4718.
7. Di Trapani F., Oddo M.C., *Sberna A.P.*, La Mendola L. "Structural health monitoring of masonry structures using stress sensors: Experimental induced damage tests and proposed approach for real-time monitoring". *Construction and Building Materials*, 2024; 449:138077.
8. *Sberna A.P.*, Demartino C., Vanzi I., Marano G.C., Di Trapani F. "Cost-effective topology optimization of masonry structure reinforcements by a linear static analysis-based GA framework". *Bulletin of Earthquake Engineering*, 2024; 22:4143–4167.
9. Di Trapani F., *Sberna A.P.*, Di Benedetto M., Villar S., Demartino C., Marano G.C. "Dynamic progressive collapse response of multi-storey frame structures with masonry infills". *Structures*, 2023; 54:1336–1349.
10. Di Trapani F., *Sberna A.P.*, Marano G.C. "A genetic algorithm-based framework for seismic retrofitting cost and expected annual loss optimization of non-conforming reinforced concrete frame structures". *Computers and Structures*, 2022; 271:106855.
11. Di Trapani F., Vizzino A., Tomaselli G., *Sberna A.P.*, Bertagnoli G. "A new empirical formulation for the out-of-plane resistance of masonry infills in reinforced concrete frames". *Engineering Structures*, 2022; 266:114422.
12. Di Trapani F., *Sberna A.P.*, Marano G.C. "A new genetic algorithm-based framework for optimized design of steel-jacketing retrofitting in shear-critical and ductility-critical RC frame structures". *Engineering Structures*, 2021; 243:112684.

13. Di Trapani F., Malavisi M., Marano G.C., *Sberna A.P.*, Greco R. "Optimal seismic retrofitting of reinforced concrete buildings by steel-jacketing using a genetic algorithm-based framework". *Engineering Structures*, 2020; 219:110864.

Manuscripts under review

1. *Sberna A.P.*, Di Trapani F. "Closed-form biaxial curvature and ductility domains for reinforced concrete rectangular sections". Submitted to *Engineering Structures*.
2. Colajanni P., Cucchiara C., Granata M.F., *Sberna A.P.* "Influence of Bedrock Seismic Action Modelling on Microzonation". Submitted to *Buildings*. Preprint available.
3. Zhou S.C., Demartino C., Xu J.J., Lai D., Wang J., *Sberna A.P.*, Xiao Y. "Near-Base Lateral Impact Response of Circular Concrete-Filled Steel Tube Bridge-Pier Columns: Experimental Comparison of Cantilever and Propped-Cantilever Boundary Conditions". Submitted to *Engineering Structures*.

Conference proceedings

1. *Sberna A.P.*, Di Trapani F. "Genetic algorithm framework for environmentally sustainable and cost-efficient seismic retrofitting design of reinforced concrete structures". *4th fib Italy YMG Symposium on Concrete and Concrete Structures*, Naples (Italy), September 2025. In press.
2. Campisi F., Di Trapani F., Di Leto M., Cucchiara C., *Sberna A.P.*, La Mendola L. "Seismic risk reduction of masonry barrel vaults using CRM: fragility assessment using a fiber-section approach". *20th ANIDIS Conference*, Assisi (Italy), September 2025. Published in *Procedia Structural Integrity*, 2026; 78:1197–1204.
3. *Sberna A.P.*, Quaranta G., Di Trapani F. "Analytical formulation for biaxial curvature domains of rectangular reinforced concrete sections". *20th ANIDIS Conference*, Assisi (Italy), September 2025. Published in *Procedia Structural Integrity*, 2026; 78:1879–1886.
4. Villar S., Di Benedetto M., *Sberna A.P.*, Petracca M., Camata G., Di Trapani F. "Seismic Response of Masonry Aggregate Buildings by Explicit Modelling Different Interconnection Degrees". *6th Eurasian Conference on OpenSees Days – EOSD 2024*, Beijing (China), July 2024. Published in *Lecture Notes in Civil Engineering*, 2025; pp. 183–192.
5. Di Benedetto M., Di Trapani F., *Sberna A.P.*, Villar S., Ferrara M., Bertagnoli G. "Numerical Modeling of Dapped-End Beams Subjected to Corrosion and Retrofitting Interventions". *6th Eurasian Conference on OpenSees Days – EOSD 2024*, Beijing (China), July 2024. Published in *Lecture Notes in Civil Engineering*, 2025; pp. 111–121.
6. Di Trapani F., *Sberna A.P.*, Bertagnoli G., Marano G.C. "A novel genetic algorithm-based optimization framework for minimizing seismic retrofitting costs in existing masonry structures". *19th ANIDIS Conference – Seismic Engineering in Italy*, Turin (Italy), September 2022.
7. *Sberna A.P.*, Di Trapani F., Marano G.C. "A new GA-based framework based on Expected Annual Loss for optimizing seismic retrofitting in reinforced concrete frame structures". *19th ANIDIS Conference – Seismic Engineering in Italy*, Turin (Italy), September 2022.
8. Di Trapani F., *Sberna A.P.*, Demartino C., Marano G.C. "Genetic algorithm-based seismic retrofitting optimization for existing masonry structures". *3ECEES – 3rd European Conference on Earthquake Engineering and Seismology*, Bucharest (Romania), September 2022.

9. Di Trapani F., Vizzino A., Tomaselli G., *Sberna A.P.*, Bertagnoli G. "Predictive equations for the out-of-plane resistance of masonry infills in RC frames by a data-driven approach". *3ECEES – 3rd European Conference on Earthquake Engineering and Seismology*, Bucharest (Romania), September 2022.
10. Di Trapani F., *Sberna A.P.*, Demartino C., Marano G.C. "Genetic algorithm-based optimization procedure for the seismic retrofitting of existing masonry structures". *CST – 14th International Conference on Computational Structures Technology*, Montpellier (France), August 2022.
11. Di Trapani F., Bogatkina V., Di Benedetto M., *Sberna A.P.*, Petracca M., Camata G. "Simplified Evaluation of the Additional Shear Demand Due to Masonry Infills". *EOSD 2022 – 2nd Eurasian Conference on OpenSees Days*, Turin (Italy), July 2022. Published in *Lecture Notes in Civil Engineering*, 2023; 326:1–13.
12. *Sberna A.P.*, Di Trapani F., Marano G.C. "A novel framework based on Expected Annual Loss for optimizing seismic retrofitting in reinforced concrete structures". *2nd Symposium on Concrete and Concrete Structures*, Rome (Italy), November 2021.
13. Di Trapani F., Tomaselli G., *Sberna A.P.*, Rosso M.M., Marano G.C. "Dynamic response of infilled frames subject to accidental column losses". *EUROSTRUCT 2021 – 1st Conference of the European Association on Quality Control of Bridges and Structures*, Padua (Italy), September 2021.
14. Di Trapani F., *Sberna A.P.*, Marano G.C. "Expected Annual Loss oriented seismic retrofitting optimization of RC frame structures using a new AI-based framework". *COMPDYN 2021 – 8th ECCOMAS International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*, Athens (Greece), June 2021.
15. *Sberna A.P.*, Di Trapani F., Marano G.C. "Optimization of steel-jacketing retrofitting of shear-critical and ductility-critical RC frame structures by a novel genetic algorithm framework". *COMPDYN 2021 – 8th ECCOMAS International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering*, Athens (Greece), June 2021.
16. Di Trapani F., *Sberna A.P.*, Marano G.C. "Cost and EAL based optimization for seismic reinforcement of RC structures". *IGF26 – 26th International Conference on Fracture and Structural Integrity*, Turin (Italy), May 2021.
17. Di Trapani F., *Sberna A.P.*, Tomaselli G., Marano G.C. "Cost-based and EAL based optimization algorithms for seismic retrofitting of RC frame structures". *Italian Concrete Days 2020*, Naples (Italy), April 2021.
18. Di Trapani F., Malavisi M., *Sberna A.P.*, Marano G.C. "Genetic algorithm-based optimization of RC frame structures retrofitting with steel jacketing". *IWSS 2020 – 1st Italian Workshop on Shell and Spatial Structures*, Turin (Italy), June 2020.

Conference special session organizer and chair

Mini symposium: "Reduced-Order and Surrogate Models: Advances in Computationally Efficient Structural Dynamics" at COMPDYN 2027, Rhodes, Greece, July 2027.

Mini symposium: "Recent Advancements in OpenSees for Earthquake Engineering: Methods, Tools, and Applications" at COMPDYN 2027, Rhodes, Greece, July 2027.

Chair: "Session 1a - Session 2" at 7th Eurasian Conference on OpenSees, Lausanne, Switzerland, July 2026.

Special Session: "Optimization-based strategies for the design and rehabilitation of sustainable concrete structures" at 7th fib Congress. Lisbon, Portugal, June 2026.

Chair: "Analysis methods, modeling and capacity models" at XX ANIDIS Conference, Seismic Engineering in Italy. Assisi, Italy, September 2025.

Special session and Chair: "Advancements in OpenSees applications for Earthquake Engineering" at XX ANIDIS Conference, Seismic Engineering in Italy. Assisi, Italy, September 2025.

Mini symposium: "Optimization-Driven Design and Retrofit Strategies for Enhancing Sustainability in Structural Systems" at COMPDYN 2025, Rhodes, Greece, June 2025.

Chair: "Session 3 - Session 6" at 6th Eurasian Conference on OpenSees, Beijing, China, July 2024.

Chair: "Session 4" at 2nd Eurasian Conference on OpenSees, Torino, Italy, July 2022.

Presentations at conferences and workshops

Sep 2026	18th European Conference on Earthquake Engineering, Berlin (Germany).
Jul 2026	7th Eurasian Conference on OpenSees, Lausanne (Switzerland).
Sep 2025	4th fib Symposium on Concrete and Concrete Structures, Naples (Italy), 15th – 16th September 2025.
Sep 2025	XX ANIDIS Conference - Italian Conference of Earthquake Engineering, Assisi (Italy), 7th – 11th September 2025.
Jun 2025	14th International Conference on Structural Safety and Reliability — ICOSSAR'25, Los Angeles (US), 1st – 6th June 2025.
Jun 2025	COMPDYN 2025 - 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Rhodes (Greece), 15th – 18th June 2025.
Jul 2024	6th Eurasian Conference on OpenSees, Beijing (China), 24th – 25th July 2024.
Jul 2024	18th World Conference of Earthquake Engineering 2024, Milan (Italy), 1st – 5th July 2024.
Aug 2023	Engineering Mechanics Institute International Conference 2023, Palermo (Italy), 27th – 30th August 2023.
Sep 2022	19th Italian Conference on Earthquake Engineering, Turin (Italy), 11th – 15th September 2022.
Sep 2022	3rd European Conference on Earthquake Engineering & Seismology, Bucharest (Romania), 4th – 9th September 2022.
Aug 2022	14th Computational Structures Technology, Montpellier (France), 23rd – 25th August 2022.
Jul 2022	2nd EOS Eurasian Conference on OpenSees, Turin (Italy), 7th – 8th July 2022.
Nov 2021	2nd FIB Symposium on Concrete and Concrete Structures, Rome (Italy), 18th – 19th November 2021.
Jun 2021	8th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Athens (Greece), 28th – 30th June 2021.

Participation in funded research projects

2024 – 2026 Research Unit Member, Politecnico di Torino / University of Chieti-Pescara, *DPC-ReLUIS 2024–2026, WP11 Reinforced Concrete Structures, Task 11.2: Methods for the Execution of Non-Linear Dynamic Analyses* – funded by Italian Civil Protection Department (DPC), coordinated by ReLUIS Consortium.

2024 – 2026 Research Unit Member, Politecnico di Torino, *DPC-ReLUIS 2024–2026, WP10 Masonry Structures, Task 10.3: Local and Global Capacity Models for the Assessment of Limit States – Non-Structural Masonry* – funded by Italian Civil Protection Department (DPC), coordinated by ReLUIS Consortium.

2024 – 2026 Adjunct (Invited) Research Unit Member, Sapienza University of Rome, *DPC-ReLUIS 2024–2026, WP11 Reinforced Concrete Structures, Task 11.1: Capacity Models* – funded by Italian Civil Protection Department (DPC), coordinated by ReLUIS Consortium.

2023 – 2025 Research Unit Member, Politecnico di Torino, *RETURN: Multi-Risk Science for Resilient Communities under a Changing Climate* – funded by European Union - NextGenerationEU, NRRP Mission 4, Component 2, Investment 1.3 (Extended Partnership PE00000005), coordinated by University of Naples Federico II.

2023 – 2025 Adjunct Research Unit Member, Politecnico di Torino, *AI-ENVISERS: Artificial Intelligence for ENVIronmental impact minimization of SEismic Retrofitting of Structures* – funded by European Union - NextGenerationEU, PRIN 2022 PNRR (MUR D.D. 1409, 14/09/2022), coordinated by Roma Tre University.

2023 – 2025 Member of the Proposing Research Unit, Politecnico di Torino, *AI-SUST: Artificial Intelligence for SUstainable seismic risk reduction of STructures* – funded by Italian Ministry of University and Research, PRIN 2022, coordinated by Politecnico di Torino.

2022 – 2026 Research Unit Member, Politecnico di Torino, *ERIES-ENFRAG, within ERIES: Engineering Research Infrastructures for European Synergies* – funded by Horizon Europe, Research Infrastructures (GA 101058684), coordinated by IUSS Pavia.

2021 – 2025 Research Unit Member, Politecnico di Torino, *ADDOPTML: ADDitively Manufactured OPTimized Structures by means of Machine Learning* – funded by H2020 MSCA-RISE-2020 (GA 101007595), coordinated by National Technical University of Athens (NTUA).

2022 – 2024 Research Unit Member, Politecnico di Torino, *DPC-ReLUIS 2022–2024, WP10 Masonry Structures, Task 10.3: Local and Global Capacity Models for the Assessment of Limit States – Non-Structural Masonry* – funded by Italian Civil Protection Department (DPC), coordinated by ReLUIS Consortium.

Member of scientific and organizing committee of international conferences and courses

Organizing committee of *7th Eurasian Conference on OpenSees* - Lausanne (Switzerland), Jul 2026.

Scientific and organizing committee of *9th International Course on Seismic Analysis of Structures using OpenSees* - Lausanne (Switzerland), 5th – 7th March 2025.

Organizing committee of *6th Eurasian Conference on OpenSees* - Beijing (China), 24th – 25th July 2024.

Scientific and organizing committee of *8th International Course on Seismic Analysis of Structures using OpenSees* - Beijing (China), 22nd – 23rd July 2024.

Scientific and organizing committee of *2nd EOS Eurasian Conference on OpenSees* - Turin (Italy), 7th – 8th July 2022.

Scientific and organizing committee of *7th International Course on Seismic Analysis of Structures using OpenSees* - Turin (Italy), 5th – 6th July 2022.

Teaching activities

Summer 2026	Lecturer, *Ph.D. Course on Nonlinear Analysis of Masonry Structures*, University of Chieti-Pescara, Italy.
Spring 2025	Teaching assistance, *Safety assessment and retrofitting of existing structures and infrastructures* (in english), graduate class - School of Engineering, Politecnico di Torino, Italy.
Spring 2024	Teaching assistance, *Safety assessment and retrofitting of existing structures and infrastructures* (in english), graduate class - School of Engineering, Politecnico di Torino, Italy.
Summer 2022	Tutor, *7th International Course on Seismic Analysis of Structures using OpenSees*, Politecnico di Torino, Italy.
Spring 2022	Teaching assistance, *Safety assessment and retrofitting of existing structures and infrastructures* (in english), graduate class - School of Engineering, Politecnico di Torino, Italy.
Summer 2021	Tutor, *6th International Course on Seismic Analysis of Structures using OpenSees*, University of Palermo, Italy.

Fellowships

MSCA Postdoctoral Fellowship – Seal of Excellence, European Commission (Awarded to a high-quality proposal that could not be funded due to budgetary constraints) - Feb 2026.

Visiting Research Scholar at the University of California, San Diego (UCSD), within the framework of the Erasmus+ Staff Exchange program - May 2025

Invited lectures and workshops

May 2026	*10th International Course on Seismic Analysis of Structures using OpenSees*, Thematic lecture on: *Dynamic analyses of structures using OpenSees*, University of Minho, Portugal.
Jun 2025	*COMPDYN 2025*, Keynote presentation on: *Genetic Algorithm-Based Framework for Environmental and Economic Sustainable Seismic Retrofitting Design of Reinforced Concrete Frame Structures*, Rhodes (Greece).
Mar 2025	*9th International Course on Seismic Analysis of Structures using OpenSees*, Thematic lecture on: *Linear and non-linear dynamic analyses of structures using OpenSees*, EPFL, Switzerland.
Jul 2024	*3rd Structural Engineering Workshop*, Palermo (Italy), 16th July 2024.
Jul 2024	*8th International Course on Seismic Analysis of Structures using OpenSees*, Thematic lecture on: *Linear and non-linear dynamic analyses of structures using OpenSees*, Tsinghua University, China.
Jan 2024	*Workshop - Recent research results using OpenSees*, University College London (UK), 11th January 2024.

Books and book chapters

Chapter 3: *Sustainable Retrofitting Toward Resilient Buildings & Infrastructures - Optimal retrofitting design* (on preorder, it will be released on 27 December 2026), Tiago Ferreira - World Scientific Publishing Co Pte Ltd, 2026.

Working group member: *Progettazione e verifica sismica di tamponature murarie in edifici in c.a.* (*in Italian*), P. Morandi, P. Ricci, G. Magenes, ReLUIS / Doppiavoce, 2026.

Di Trapani F., *Structural Analysis Methods: Beam and Planar Frame Systems*. Chapter 6 - Performing planar frame structures using MATLAB®, CLUT Editore, Turin 2022. ISBN: 9788879924894

Thesis advisor

Riccardo Ditta - *Assessment of aleatory and epistemic uncertainties in the seismic response of reinforced concrete structures with masonry infill walls*. Supervisors: Di Trapani F., Taner Y., Sberna A.P., University of Palermo, 2025.

Luca Biancotto - *Fragility functions of unreinforced masonry aggregate buildings under varying unit-interconnectivity conditions*. Supervisors: Castaldo P., Di Trapani F., Villar S., Sberna A.P., Politecnico di Torino, 2025.

Fabio Fraschini - *Development of a Graphical User Interface for the Analysis of Reinforced Concrete Sections*. Supervisors: Di Trapani F., Sberna A.P., Politecnico di Torino, 2024.

Cristian Marra - *Definition of reduced structural models for the nonlinear dynamic analysis of framed reinforced concrete structures*. Supervisors: Di Trapani F., Sberna A.P., Politecnico di Torino, 2024.

Santosh Shrestha - *Calibration of a multi-strut macro model for seismic analysis of infilled reinforced concrete frames*. Supervisors: Di Trapani F., Sberna A.P., Di Benedetto M., Politecnico di Torino, 2024.

Alexandre Milliet - *Investigation of collapse resistance in tall timber buildings*. Supervisors: Frangi A., Cao A.S., *Sberna A.P.*, ETH Zurich, 2023.

Dario Coletta - *Optimization of seismic reinforcement for masonry structures using genetic algorithms*. Supervisors: Di Trapani F., *Sberna A.P.*, Politecnico di Torino, 2021.

Reviewer for international journals

Structural Engineering

Engineering Structures

Computers & Structures

Structures

Natural Hazards

Reliability Engineering & System Safety

Structural Concrete

Journal of Building Engineering

Construction and Building Materials

Journal of Asian Architecture and Building Engineering

Discover Applied Sciences

Advances in Civil Engineering

Buildings

Journal of Brazilian Society of Mechanical Sciences and Engineering

Sustainability

Modelling

Infrastructure

Archives of Computational Methods in Engineering

Structures and Buildings (Proceedings of ICE)

Smart Infrastructures and Construction (Proceedings of ICE)

Engineering History and Heritage (Proceedings of ICE)

Engineering and Computational Mechanics (Proceedings of ICE)

Transport (Proceedings of ICE)

Forensic Engineering

Journal of Structural Integrity and Maintenance

Advances in Structural Engineering

Algorithms, Optimization and AI

Engineering Applications of Artificial Intelligence

Algorithms

AI

Symmetry

Drones

Big Data and Cognitive Computing

Biomimetics

Applied Artificial Intelligence

Proceedings

13th U.S. National Conference on Earthquake Engineering

20th Italian Conference on Earthquake Engineering

6th Eurasian Conference on OpenSees

19th Italian Conference on Earthquake Engineering

2nd EOS Eurasian Conference on OpenSees

Pescara (Italy); November 10th, 2026 *Antonio P. Sberna*